Q1. JEE Main 2026 (21 January Shift 2)

Let α and β be the roots of the equation $x^2+2ax+(3a+10)=0$ such that $\alpha<1<\beta$. Then the set of all possible values of a is :

(1) $\left(-\infty, \frac{-11}{5}\right)$
(2) $(-\infty,-2) \cup(5, \infty)$
(3) $(-\infty,-3)$
(4) $\left(-\infty, \frac{-11}{5}\right) \cup(5, \infty)$

Q2. JEE Main 2026 (23 January Shift 1)

A building construction work can be completed by two masons A and B together in 22.5 days. Mason A alone can complete the construction work in 24 days less than mason B alone. Then mason A alone will complete the construction work in :

(1) 30 days
(2) 36 days
(3) 24 days
(4) 42 days

Q3. JEE Main 2026 (23 January Shift 1)

If α and $\beta(\alpha<\beta)$ are the roots of the equation $(-2+\sqrt{3})(|\sqrt{x}-3|)+(x-6\sqrt{x})+(9-2\sqrt{3})=0, x \geqslant 0$, then $\sqrt{\frac{\beta}{\alpha}}+\sqrt{\alpha \beta}$ is equal to :

(1) 8
(2) 11
(3) 9
(4) 10

Q4. JEE Main 2026 (24 January Shift 2)

The smallest positive integral value of a, for which all the roots of $x^4-ax^2+9=0$ are real and distinct, is equal to

(1) 4
(2) 9
(3) 3
(4) 7

Q5. JEE Main 2026 (22 January Shift 2)

Let α, β be the roots of the quadratic equation $12x^2-20x+3\lambda=0, \lambda \in \mathbf{Z}$. If $\frac{1}{2} \leqslant|\beta-\alpha| \leqslant \frac{3}{2}$, then the sum of all possible values of λ is :

(1) 1
(2) 6
(3) 3
(4) 4

Q6. JEE Main 2026 (23 January Shift 2)

The sum of all the real solutions of the equation
$\log_{(x+3)}\left(6x^2+28x+30\right)=5-2\log_{(6x+10)}\left(x^2+6x+9\right)$ is equal to

(1) 2
(2) 1
(3) 0
(4) 4

Q7. JEE Main 2026 (21 January Shift 1)

The sum of all the roots of the equation $(x-1)^2-5|x-1|+6=0$, is :

(1) 1
(2) 5
(3) 3
(4) 4

Q8. JEE Main 2026 (28 January Shift 1)

Let S $= \{x^3 + ax^2 + bx + c : a, b, c \in$ N and $a, b, c \leq 20\}$ be a set of polynomials. Then the number of polynomials in S, which are divisible by $x^2 + 2$, is

(1) 10 (2) 20 (3) 6 (4) 120

Q9. JEE Main 2026 (28 January Shift 1)

If α, β, where $\alpha < \beta$, are the roots of the equation $\lambda x^2 - (\lambda + 3)x + 3 = 0$ such that $\frac{1}{\alpha} - \frac{1}{\beta} = \frac{1}{3}$, then the sum of all possible values of λ is

(1) 8 (2) 6 (3) 2 (4) 4

ANSWERS AND SOLUTIONS

1. (1) **2.** (2) **3.** (4) **4.** (4) **5.** (3) **6.** (3) **7.** (4) **8.** (1)
9. (2)

1. **(1)** Let $f(x) = x^2 + 2ax + (3a + 10)$. For $\alpha < 1 < \beta$, since the leading coefficient is positive, we need $f(1) < 0$.
$f(1) = 1 + 2a + 3a + 10 = 5a + 11 < 0$
$a < -\frac{11}{5}$
When $f(1) < 0$ with positive leading coefficient, the parabola is negative at $x = 1$, so 1 lies strictly between the two real roots. Discriminant is automatically positive.
$a \in \left(-\infty, -\frac{11}{5}\right)$.

2. **(2)** Let A take a days and B take $a + 24$ days.
Together: $\frac{1}{a} + \frac{1}{a+24} = \frac{1}{22.5} = \frac{2}{45}$.
$\frac{2a+24}{a(a+24)} = \frac{2}{45}$
$45(2a + 24) = 2a(a + 24)$
$90a + 1080 = 2a^2 + 48a$
$a^2 - 21a - 540 = 0$
$(a - 36)(a + 15) = 0 \Rightarrow a = 36$.
Mason A alone completes the work in 36 days.

3. **(4)** Let $t = \sqrt{x} \geq 0$. The equation becomes $(-2 + \sqrt{3})|t - 3| + (t^2 - 6t) + (9 - 2\sqrt{3}) = 0$.
Since $t^2 - 6t + 9 = (t - 3)^2$, we get $(t - 3)^2 + (\sqrt{3} - 2)|t - 3| - 2\sqrt{3} = 0$.
Let $u = |t - 3| \geq 0$: $u^2 + (\sqrt{3} - 2)u - 2\sqrt{3} = 0$.
Discriminant $= (\sqrt{3} - 2)^2 + 8\sqrt{3} = 7 + 4\sqrt{3} = (2 + \sqrt{3})^2$.
$u = \frac{(2-\sqrt{3})\pm(2+\sqrt{3})}{2}$, giving $u = 2$ or $u = -\sqrt{3}$ (rejected).
$|t - 3| = 2 \Rightarrow t = 5$ or $t = 1$, so $x = 25$ or $x = 1$.
$\alpha = 1, \beta = 25$.
$\sqrt{\beta/\alpha} + \sqrt{\alpha\beta} = \sqrt{25} + \sqrt{25} = 5 + 5 = 10$.

4. **(4)** Let $t = x^2$, so the equation becomes $t^2 - at + 9 = 0$.
For all four roots of the original equation to be real and distinct, both roots of this quadratic must be positive and distinct.
Discriminant > 0: $a^2 - 36 > 0 \Rightarrow |a| > 6$. Since $a > 0$, we need $a > 6$.
Product of roots $= 9 > 0$ and sum of roots $= a > 0$, so both roots are positive.
The smallest positive integer satisfying $a > 6$ is $a = 7$.

5. **(3)** For $12x^2 - 20x + 3\lambda = 0$: $\alpha + \beta = \frac{5}{3}$, $\alpha\beta = \frac{\lambda}{4}$.
$|\beta - \alpha| = \frac{\sqrt{25-9\lambda}}{3}$

Given $\frac{1}{2} \leq \frac{\sqrt{25-9\lambda}}{3} \leq \frac{3}{2}$

$\frac{9}{4} \leq 25 - 9\lambda \leq \frac{81}{4}$

From left: $\lambda \leq \frac{91}{36} \approx 2.53$

From right: $\lambda \geq \frac{19}{36} \approx 0.53$

Since $\lambda \in \mathbb{Z}$: $\lambda \in \{1, 2\}$. Sum $= 3$.

6. **(3)** Note $6x^2 + 28x + 30 = 2(3x+5)(x+3)$,

$x^2 + 6x + 9 = (x+3)^2$,

$6x + 10 = 2(3x+5)$.

Let $a = \log_{(x+3)}[2(3x+5)]$. Then LHS $= a + 1$ and $\log_{2(3x+5)}(x+3)^2 = \frac{2}{a}$.

Equation becomes $a + 1 = 5 - \frac{4}{a}$, i.e., $a^2 - 4a + 4 = 0 \Rightarrow a = 2$.

So $\log_{(x+3)}[2(3x+5)] = 2 \Rightarrow (x+3)^2 = 6x + 10 \Rightarrow x^2 = 1 \Rightarrow x = 1$ or $x = -1$.

Both satisfy domain conditions (bases $> 0, \neq 1$; arguments > 0).

Sum $= 1 + (-1) = 0$.

7. **(4)** Let $t = |x - 1|$ where $t \geq 0$.

Equation becomes $t^2 - 5t + 6 = 0$.

$(t-2)(t-3) = 0$ gives $t = 2$ or $t = 3$.

For $|x - 1| = 2$: $x = 3$ or $x = -1$.

For $|x - 1| = 3$: $x = 4$ or $x = -2$.

Sum of roots $= 3 + (-1) + 4 + (-2) = 4$.

8. **(1)** If $x^2 + 2$ divides $x^3 + ax^2 + bx + c$, write $x^3 + ax^2 + bx + c = (x^2 + 2)(x + k)$.

Expanding: $x^3 + kx^2 + 2x + 2k$.

Comparing coefficients: $a = k$, $b = 2$, $c = 2a$.

Constraints: $a \in \mathbb{N}$, $a \leq 20$, and $c = 2a \leq 20 \Rightarrow a \leq 10$.

So $a \in \{1, 2, 3, \ldots, 10\}$, giving 10 polynomials.

9. **(2)** $\alpha + \beta = \frac{\lambda+3}{\lambda}$, $\alpha\beta = \frac{3}{\lambda}$.

$\frac{1}{\alpha} - \frac{1}{\beta} = \frac{\beta-\alpha}{\alpha\beta} = \frac{1}{3}$, so $\beta - \alpha = \frac{\alpha\beta}{3} = \frac{1}{\lambda}$.

$(\beta - \alpha)^2 = (\alpha + \beta)^2 - 4\alpha\beta$:

$\frac{1}{\lambda^2} = \frac{(\lambda+3)^2}{\lambda^2} - \frac{12}{\lambda}$

$1 = (\lambda + 3)^2 - 12\lambda = \lambda^2 - 6\lambda + 9$

$\lambda^2 - 6\lambda + 8 = 0 \Rightarrow \lambda = 2$ or $\lambda = 4$.

Both values satisfy $\alpha < \beta$ (verified). Sum $= 2 + 4 = 6$.